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# Mitigation of Gap Losses in Nanocrystalline Tape-Wound Cores

Gerardo Calderon-Lopez, Yiren Wang and Andrew J. Forsyth, *Senior Member, IEEE*

**Abstract**— A split core technique is proposed to mitigate the gap losses in high frequency nanocrystalline cores, which enables significant size reductions in thermally limited designs. Finite element analysis is used to examine the gap loss dependency on core width,  $D$ , revealing a non-linear relationship of the form  $\text{loss} \propto D^\alpha$ .  $\alpha$  is approximately constant for frequencies of 10 kHz – 200 kHz over the range of core widths typically used in power electronics, but  $\alpha$  increases with gap length. Splitting the core into a number of sub-cores can therefore provide significant reductions in gap loss, especially with larger gap lengths. The results from a 300 A (peak), 200 A (continuous) inductor show that with three sub-cores and a gap length of 4 mm, the gap losses are reduced by 50%, and the hot-spot temperature is reduced by 24.5 °C. Using the technique it is estimated that the original inductor weight could be reduced by 40% with four split-cores, making a significant impact on converter power density.

**Index Terms**—Gap losses, gapped high-frequency inductor, nanocrystalline tape-wound core.

## I. INTRODUCTION

ONE of the obstacles to increasing the power density of multi-kW DC-DC converters is the size of the DC inductors, which can account for 30-50% of the overall converter weight [1, 2]. Recent technical advances are enabling reductions in inductor size, for example through increased operating frequency using SiC devices, also through increased DC and AC flux densities using new low-loss materials such as nanocrystalline tape-wound cores, where the saturation level is around 1 T and the core losses are similar to those in ferrite. Nevertheless, minimizing losses and limiting the hot-spot temperature within the component remain key challenges in reducing inductor size.

Gap losses arise in high current DC inductors using nanocrystalline cores from the requirement to include an air gap in the magnetic circuit. The component of the air-gap fringe field that is normal to the lamination layers induces

eddy currents in the outermost laminations, which can cause intense heating at the gap edges, Fig. 1. This is in addition to conventional eddy current losses in the core. Splitting the air gap into multiple smaller gaps is one way to reduce the losses, however the machining is difficult due to the extremely thin laminations (typically 18-25  $\mu\text{m}$ ) and the hardness of the material, and increased core losses have been reported in such structures [3]. It is believed that the cutting process can produce short-circuits between laminations [3-6], and localized annealing.

Alternative methods to combat gap losses include the attachment of ferrite plates across the core area [7, 8], creating narrow slits near the gap [9], and beveling or shaping the core edges at the air gap [10, 11]. However, the use of ferrite plates can lower the saturation point of the nanocrystalline material [7], whereas the slitting/beveling can create short circuits between laminations, potentially increasing the losses. Gap losses may be eliminated by using different cores, for example with ferrites the high resistivity prevents eddy currents, however the flux must be limited to below 0.5 T, alternatively with powder iron cores the distributed gap nature of the material eliminates the need for a discrete gap, but the inductance becomes dependent on current, and the core losses are higher.

The long-established empirical expression for the calculation of gap losses suggests that the loss increases linearly with the core width [12-14], dimension  $D$  in Fig. 1, however, recent work to model the loss mechanism using finite element analysis (FEA) has suggested a non-linear relation exists between core width and gap loss [15]. Other work has also used FEA to examine gap loss, but only at frequencies up to 10 kHz [16, 17]. In this paper FEA techniques are used to examine the behavior of gap losses across a broad range of design parameters, revealing that the

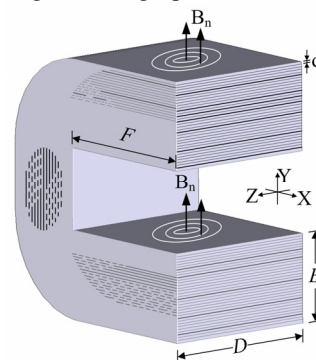


Fig. 1. In-plane eddy currents in one half of a laminated nanocrystalline U-core with air gaps.

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G. Calderon-Lopez and A. J. Forsyth are with the School of Electrical and Electronic Engineering, The University of Manchester, Manchester, M13 9PL, United Kingdom (e-mail: gerardo.calderon-lopez@manchester.ac.uk, andrew.forsyth@manchester.ac.uk).

Y. Wang is with the Hardware Department, ZYTEK Automotive/Continental Engineering Services, WS13 8RY, United Kingdom (e-mail: yiren.wang@conti-engineering.com).

losses may be significantly reduced by splitting the core into a number of sub-cores. This important new finding is validated on a prototype inductor using three split cores, showing a reduction in gap loss of approximately 50%. The importance of the technique is highlighted by a design study on a high current inductor showing a weight reduction of 40%.

Compared to our previous work, this paper presents a more extensive examination of gap loss variation with design parameters, particularly highlighting the non-linear relationship with core width and gap length. However, the key contribution is to show how splitting the core into a number of sub-cores can significantly reduce gap losses and enable reduced weight components in thermally limited designs.

## II. GAP LOSS MODELLING AND THE SPLIT CORE APPROACH

The inductor under consideration consists of tape-wound U-cores, Fig. 2. Each leg has an air gap of  $l_g/2$  and is wound with copper foil. The windings are connected in series and the entire assembly is potted in an aluminum can using a thermally conductive epoxy. Normally, the core halves would consist of a single U-core, however, to mitigate the gap losses, a split core structure is investigated here, where, as an example, the core in Fig. 2 comprises three identical cores stacked together. The stacked cores have the same effective area, and produce the same inductance as a single core with the same number of turns and gap length.

To examine the gap loss distribution in the split core structure, 3-D FEA simulations were undertaken in Opera 3-D. The finely laminated core was modelled using the homogenization method described in [15], whereby the core is represented by a continuum of material having anisotropic properties. This avoids having to mesh the individual laminations and simplifies the computation. Linear magnetic properties were assumed, and the solver used time-harmonic equations.

Table I summarizes the anisotropic properties of the homogenized core model, assuming Finemet material, which has a measured differential permeability of 2500 for a flux density up to 1 T, an electrical conductivity of  $8.33 \times 10^5$  S/m [15], and a core packing factor of 0.78. The normal and tangential directions of the parameters are with respect to the lamination layers. Using the effective core parameters in Table I, the skin depth in the core at say 50 kHz and 200 kHz is 1.23 mm and 0.62 mm, respectively. To ensure accurate calculation of the eddy currents in the core, the mesh size was set to be 0.2 mm. The windings were used only as magnetic sources in the model and they were represented by current-driven Biot-Savart conductors with sinusoidal excitation. The additional losses generated by the fringing flux in the surrounding conductors were not considered.

### A. Gap Loss Variation with Core Width, Gap Length and Frequency

A range of core widths, gap lengths and excitation frequencies were modelled, whilst the core thickness and limb length,  $E$  and  $F$  in Fig. 1, and the average flux density,  $B_m$ , were kept constant. Core width, gap length and frequency

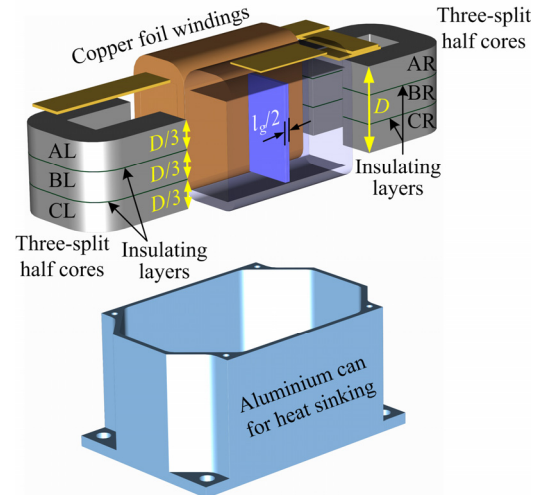


Fig. 2. High current DC inductor with three-split cores for gap loss mitigation.

TABLE I  
ANISOTROPIC PROPERTIES OF THE HOMOGENIZED CORE MODEL

| Parameter               | Symbol     | Value              | Unit |
|-------------------------|------------|--------------------|------|
| Normal conductivity     | $\sigma_n$ | 0.46               | S/m  |
| Normal permeability     | $\mu_n$    | 5                  | —    |
| Tangential conductivity | $\sigma_t$ | $6.67 \times 10^5$ | S/m  |
| Tangential permeability | $\mu_t$    | 2000               | —    |

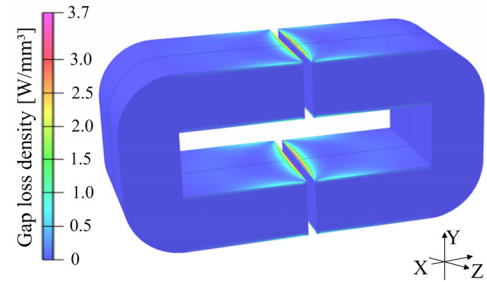


Fig. 3. Gap loss distribution in the core.

were identified from a large number of simulations to be the most significant in determining the gap losses; however, the effect of the other parameters is discussed briefly in Section B. Fig. 3 shows an example loss density plot, where the losses are seen to be concentrated mainly at the gap edge on the core surfaces and to some extent down the sides of the core. By integrating the loss density over the entire limb of the core, the total eddy current loss was obtained. This calculation includes the effects of conventionally eddy current losses, however, by analyzing an ungapped core with the same average flux density, the total eddy current losses were found to be a small fraction of those in the gapped cores, typically less than 5%, therefore the total eddy current loss in the gapped cores was assumed to provide a good approximation for the gap loss.

Fig. 4 shows the variation of gap loss with core width at 2 kHz and 30 kHz. For the larger core widths the lines are approximately straight at all frequencies, but the gradients are larger at higher frequency and with larger gap lengths. That is for a fixed frequency and gap length, the marginal increase in loss with core width is linear. However, the gradients of the lines reduce as the core width becomes small, with the loss tending to zero for very small widths. This is thought to be caused by very narrow cores preventing the flow of significant

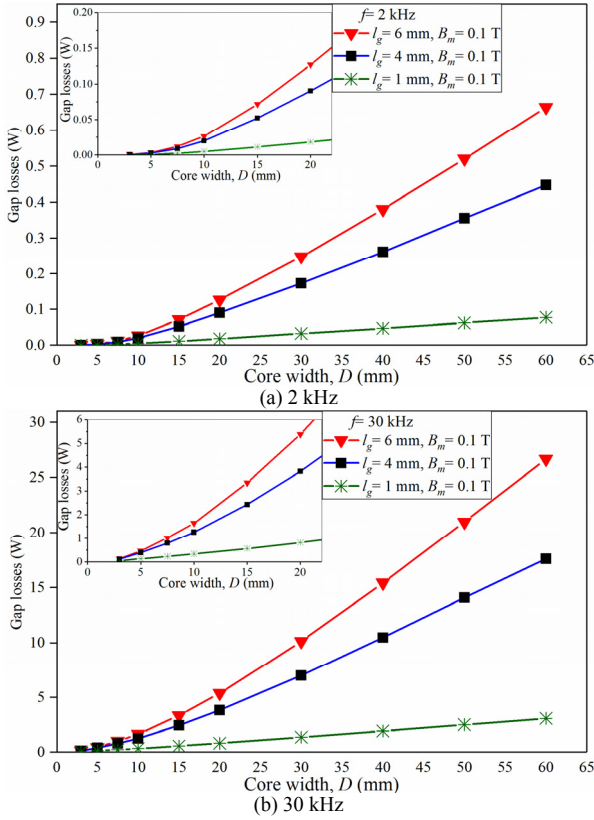


Fig. 4. Gap loss variation with core width,  $E = 13$  mm,  $F = 28$  mm: (a) 2 kHz and (b) 30 kHz. Inset plot shows a magnified view of the region  $3 \leq D < 22$ .

gap loss eddy currents. From the inset plots in Fig. 4 showing the region close to the origin, the loss at 2 kHz has reduced to almost zero for a width of 5 mm, which is just less than one skin depth (6.2 mm) in the homogenized core model at this frequency. Whereas at 30 kHz the losses are still evident with a width of 5 mm, which is around three skin depths (1.6 mm). The plots also show that increasing the gap length increases the loss significantly, but does not change the overall shape of the curves. The increased loss with increased gap length is well known and is attributed to the variation of the fringe field; the field is more concentrated around the gap for small gap lengths but spreads further from the gap with larger gaps.

Comprehensive FEA results for the gap loss are shown in Fig. 5 using  $\log\text{-}\log$  plots for a wide range of frequencies, gap lengths and core widths, which cover the typical applications of nanocrystalline cores in power electronic converters. The plots are seen to form approximately straight lines for frequencies above 10 kHz, suggesting that a simple power relationship in which the gap loss is proportional to  $D^\alpha$  is a good approximation to the data, where  $\alpha$  is the gradient of the  $\log\text{-}\log$  plot. However, for frequencies below 10 kHz and for smaller core widths a simple power approximation to the data will be inaccurate and would over predict the loss. Furthermore, comparing the plots in Fig. 5, it is seen that the gradients increase with gap length, which is consistent with the relationship in Fig. 4. Therefore the exponent  $\alpha$  in the simple approximation for gap loss ( $\text{loss} \propto D^\alpha$ ) increases from  $\alpha = 1.3$  with  $l_g = 1$  mm to  $\alpha = 1.8$  with  $l_g = 6$  mm.

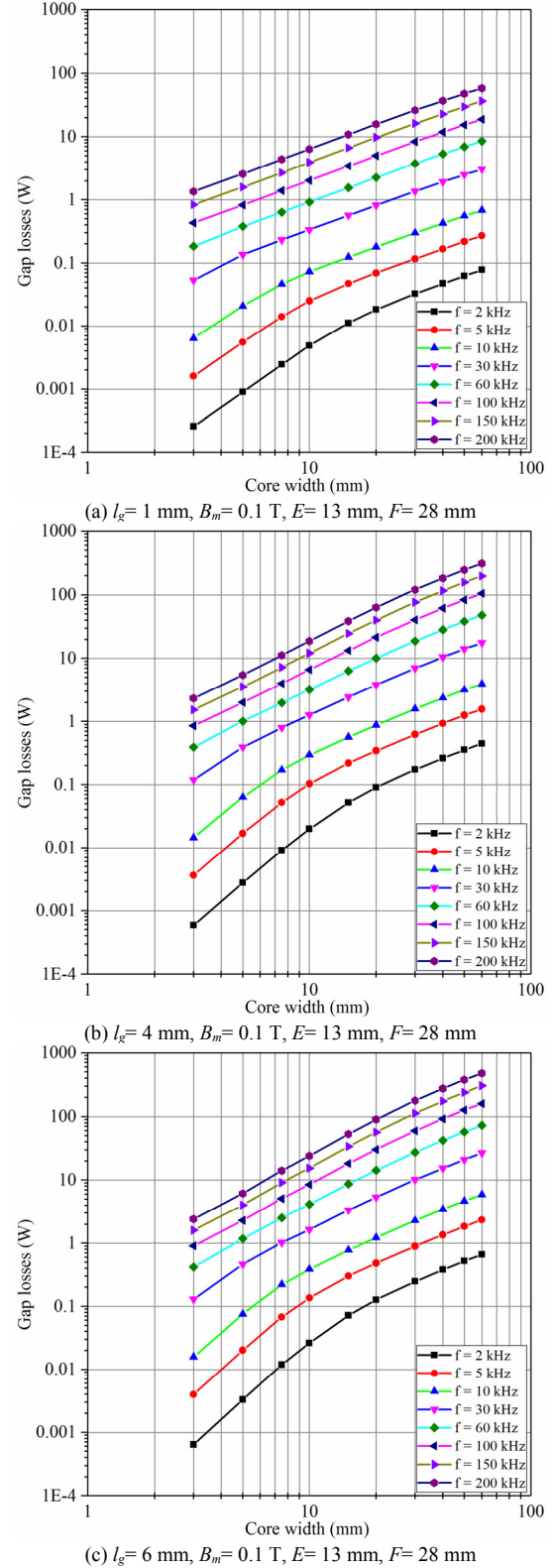


Fig. 5. Variation of gap losses with the core width at different frequencies and gap lengths.

The results confirm the general form of the gap loss approximation formula presented in [15], but clarify the frequency range over which the formula is applicable and also highlight that the exponent  $\alpha$  for the core width is dependent on the gap length.



### B. Gap Loss Sensitivity to other Parameters

Whilst the most pertinent influences on gap loss are discussed in Section A, the effects of other parameters have been examined through extensive FEA simulations. Over all the operating conditions that have been considered, it was found that the gap loss is proportional to the square of the AC flux density, which is consistent with the original empirical gap loss model [12-15]. The thickness of the core limbs, dimension  $E$ , Fig. 1, was found to have no significant effect on the gap loss and this is consistent with the loss mechanism being a surface effect as indicated in Fig. 3. The gap losses were found to increase with the length of the core limbs, dimension  $F$  in Fig. 1. This is attributed to the larger limb length providing an increased area for the fringe field and an increased conduction area for the eddy currents. Preliminary simulation results suggest that with a specific core limb length, the variation of gap loss with the core width and gap length is similar to that shown in Fig. 4 and Fig. 5.

### C. Gap Loss Mitigation with Split Cores

The gap loss data in Fig. 5 was used to examine the reduction in gap loss that may be achieved by replacing a single pair of U cores with several split cores placed side-by-side, but having the same magnetic volume as illustrated in Fig. 2. Equal gap losses were assumed in each sub-core of the complete core assembly. The total gap loss with a split core structure is shown in Fig. 6, for a frequency of 100 kHz with different numbers of sub-cores and different gap lengths. The data is normalized to the loss with a single core.

Fig. 6 shows that splitting the core can result in large reductions in the total loss, even with relatively few sub-cores. For example with an air gap length of 3 mm, the total loss is approximately halved with three split cores. Furthermore, much larger reductions are achieved when the air gap is greater. In contrast the loss reduction is lower with a smaller air gap, just under 30% with three split cores. The graph in Fig. 6 is almost identical for frequencies in the range 20 kHz – 200 kHz, which is due to the virtually straight and parallel characteristics in the  $\log\text{-}\log$  plot of Fig. 5.

Due to the approximately straight lines formed by the loss data in Fig. 5, suggesting that the loss is approximately proportional to  $D^\alpha$ , where  $\alpha$  is the line gradient for the frequency and gap length under consideration, then the reduction in loss by splitting the core into  $n_c$  sub-cores may be estimated as  $n_c^{(1-\alpha)}$ . However, as seen in Fig. 5, this will become inaccurate for smaller sub-core widths at lower frequencies, less than 10 mm and 30 kHz.

To confirm the predicted reduction in gap loss with multiple split cores, Fig. 7 presents the FEA gap loss density distributions with one, three and five sub-cores. The simulations assumed electric insulator boundaries between the sub-cores. Each core operates with the same frequency, 150 kHz, flux density, 0.13 T and air gap length, 4 mm. The results show identical loss distributions in each sub-core and significantly reduced loss densities when the core is split. The overall gap losses in each core assembly were calculated to be 139 W, 70 W, and 51 W, consistent with the predictions in

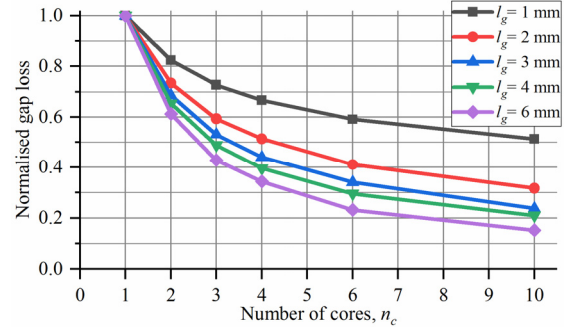


Fig. 6. Predicted gap loss from 3D FE with the split core approach with different gap lengths normalized to an equivalent single core, 100 kHz,  $D=30$  mm,  $E=13$  mm,  $F=28$  mm.

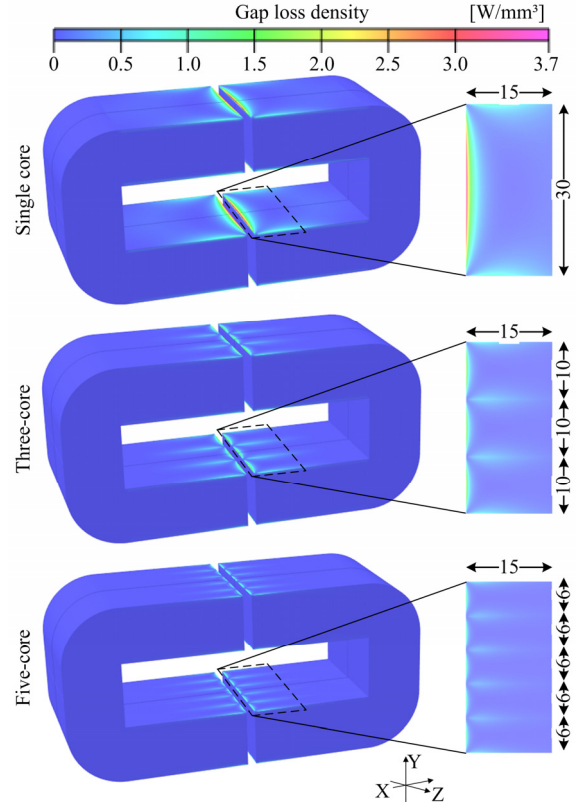


Fig. 7. Comparison of gap loss densities from FEA for single and split core structures;  $l_g=4$  mm,  $f=150$  kHz,  $B_m=0.13$  T,  $D=30$  mm,  $E=13$  mm,  $F=28$  mm.

Fig. 6 based on the simulation of individual cores.

Also the flux density level and distribution across the overall core cross section was examined in the single and split core structures at a number of points around the magnetic circuit and found to be virtually identical, apart from in the outer lamination surfaces near the gap. The conventional core losses in the single and split cores were analyzed by FEA and found to be virtually identical.

### III. EXPERIMENTAL VALIDATION

The proposed split core technique was experimentally validated with a 300 A (peak), 200 A (continuous) inductor designed to operate in a 60 kW dual-interleaved DC-DC converter with an interphase transformer [2]. The inductor used a F3CC0032 Finemet core [18], the details are listed in Table II and Fig. 8 shows how the number of turns was

TABLE II  
FEATURES OF THE SINGLE-CORE AND SPLIT-CORE INDUCTORS

| Parameter, symbol (unit)                             | Single-core | Split-core |
|------------------------------------------------------|-------------|------------|
| Number of cores, $n_c$ (-)                           | 1           | 3          |
| Overall core weight, (kg)                            | 0.460       | 0.495      |
| Overall coil weight, (kg)                            | 0.261       | 0.257      |
| Strip core width, $D$ (mm)                           | 29.9        | 10.2       |
| Thickness of core limbs, $E$ (mm)                    | 13.3        | 13.4       |
| Length of core limbs, $F$ (mm)                       | 28.1        | 28.2       |
| Number of turns, $N$ (-)                             | 6           | 6          |
| Overall gap length, $l_g$ (mm)                       | 4           | 4          |
| Copper foil cross-sectional area, (mm <sup>2</sup> ) | 30          | 30         |
| Measured potted inductance, $L_{in}$ (μH) @ 146 kHz  | 5.23        | 5.46       |
| Measured DC resistance, $R_{DC}$ (mΩ)                | 0.81        | 0.84       |
| Measured series resistance, $R_s$ (mΩ) @ 146 kHz     | 69.2        | 78.9       |

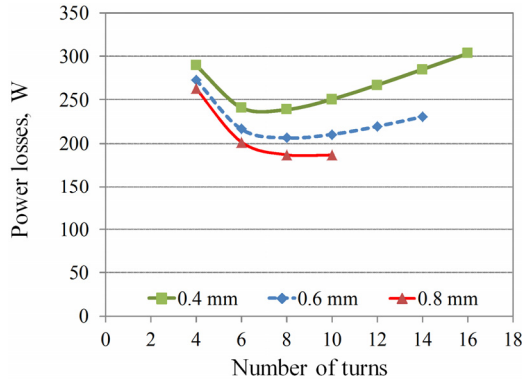


Fig. 8. Total losses for optimization of single-core inductor. F3CC0032 core with effective area 304 mm<sup>2</sup>, 300 A<sub>DC</sub>, maximum flux density 1.1 T, peak-to-peak volt-seconds 0.4 mVs, frequency 150 kHz.

determined. Fig. 8 plots the predicted inductor loss against number of turns with 0.4, 0.6 and 0.8 mm foil windings. Only even numbers of turns are considered since the winding is split between the two limbs of the core. Six turns of 0.6 mm foil were selected as this almost results in minimum loss. Slightly lower losses could have been achieved with 8 turns or 0.8 mm foil, but at the expense of increased weight. The 0.4 mm foil was rejected since the higher losses would result in an excessive temperature rise.

The original inductor was re-made using three sub-cores, each being one third of the width of the original core such that the overall magnetic volume of the two inductors was unchanged, Table II. The split-core inductor was not optimized as it was for comparison with the single-core inductor. The split core inductor used three custom cores manufactured by MK Magnetics Inc. [19] using Finemet FT-3 from Hitachi Metals with 18-μm ribbon thickness. The packing factor is the same for the single and split-cores, 0.78. The sub-cores were insulated from each other using 127 μm Kapton tape. The inductors had identical foil windings which were spaced away from the core by about the gap length per limb to minimize the fringe field losses in the winding. The inductors were each potted in an aluminum can, the thermal conductivity of the potting material being 2.1 W/mK. The AC resistance measurements were undertaken using an Agilent 4294 impedance analyzer with the core present. The contribution of the core-related losses to the measured series resistance of the winding was removed using the techniques in [20]. The effective series resistance of the core-related losses

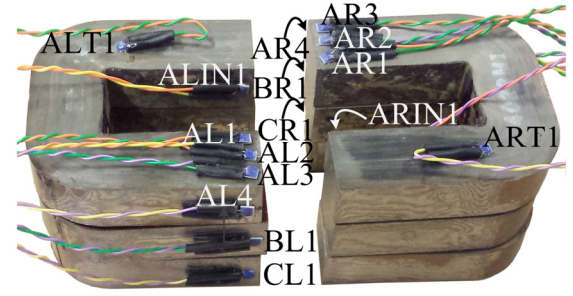


Fig. 9. Location and tags of the temperature sensors.

under the low excitation levels of the impedance analyzer was determined using the manufacturer's complex permeability data for the core and also by making separate measurements on identical cores wound with bi-filar, fine-wire windings. The second winding was connected to the voltage sense terminals of the impedance analyzer to remove the effects of winding resistance and enable direct measurement of the core loss resistance.

To confirm the expected gap loss reduction, a number of temperature sensors were fixed to the cores, especially near the gap where the hot-spots are expected to occur. Fig. 9 shows the position of the sensors for the split-core. 100-Ω RTD sensors were used due to their superior immunity to magnetic fields. Thermocouples were also embedded in the middle of the windings.

#### A. Performance Evaluation with a Calibrated Heat Sink

The inductors each operated in the DC-DC converter with identical waveforms, confirming their electrical equivalence. The inductors were mounted on a calibrated heat sink, as shown in Fig. 10, with a measured thermal resistance of 0.12 °C/W.

The heat sink was calibrated by using a resistive load with the same foot print as the potted inductor. The resistor and the top side of the heat sink were covered with thermal insulation to ensure all the heat flow was through the fins. By measuring the heat-sink steady-state temperature at several places with a range of different DC powers in the resistors, the thermal resistance of the heat sink to ambient was calculated and an average taken from the various measurements, giving the value 0.12 °C/W. The overall losses in the inductor were determined from the measured heat sink and ambient temperatures using  $P_{loss} = (T_{hsk} - T_{amb})/0.12$ .

To measure the inductor losses, the potted component, Fig. 10, was covered with thermal insulation to ensure heat-removal was solely through the heat sink. To ensure steady-state temperature conditions, the tests were run at continuous power for at least 30 minutes. Data loggers recorded the temperature rise of all the monitored spots, including the calibrated heat sink and ambient.

Several operating states at high DC current and different AC flux excitations were tested in the inductors to produce low, mid and high gap loss conditions. By changing the converter switching frequency, the inductor was operated at 148 kHz and 180 kHz. Table III summarizes the test conditions.

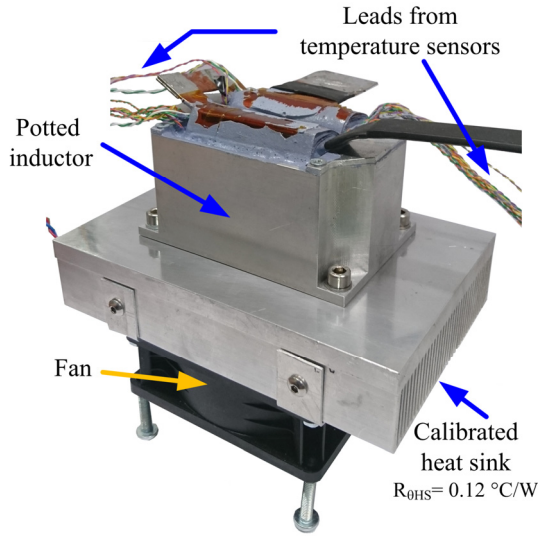


Fig. 10. Potted inductor mounted on the calibrated heat sink.

TABLE III  
OPERATING CONDITIONS IN THE DC-DC CONVERTER AND PROTOTYPE INDUCTORS

| Cond. No. | $P_{in}$ (kW) | $I_{Lin,DC}$ (A) | $B_{DC}$ (T) | $\Delta I_{Lin}$ (A <sub>pp</sub> ) | $f$ (kHz) | $B_m$ (T) | Remarks          |
|-----------|---------------|------------------|--------------|-------------------------------------|-----------|-----------|------------------|
| 1         | 53.6          | 160              | 0.51         | 45                                  | 148       | 0.068     | Low gap loss     |
| 2         | 48            | 160              | 0.51         | 76                                  | 148       | 0.11      | Medium gap loss  |
| 3         | 45.6          | 160              | 0.51         | 86                                  | 148       | 0.13      | High gap loss    |
| 4         | 42            | 150              | 0.48         | 58                                  | 180       | 0.086     | Higher frequency |

Fig. 11 displays the losses for the four operating conditions: the first two columns show the breakdown of predicted and total measured losses determined from the temperature rise in the calibrated heat sink for the single core inductor, whilst the third and fourth columns provide the predicted and measured losses for the split-core inductor. The core losses were calculated from the manufacturer's data (which include hysteresis, classical and anomalous eddy currents) using the Natural Steinmetz Extension [21], and the estimated copper losses included the skin and proximity effect [22]. Air gap fringe field related losses in the windings were not included due to the spacing of the windings away from the core, the spacing was 2.9 mm. Established analysis of fringe-field losses in foil windings [23, 24] suggest that this would under predict the copper losses by up to 14 % in our inductors, corresponding to an error in the overall loss prediction of around 3 %. The predicted core and copper losses are assumed identical in the single and split-core inductors. The FEA simulations in Section II.C confirm identical flux density distributions and conventional core losses in the single and split cores. The fields outside the core that impinge on the winding were also examined by FEA simulation and seen to be identical for the different core structures, which justifies the assumption of identical eddy current losses in the windings. Furthermore, Table II shows that the two components have very similar values of DC and AC resistance, which also confirms that they will have virtually identical copper losses. The predicted AC/DC resistance ratio at 146 kHz, using [22], is 84.4 which is very close to the measured values.

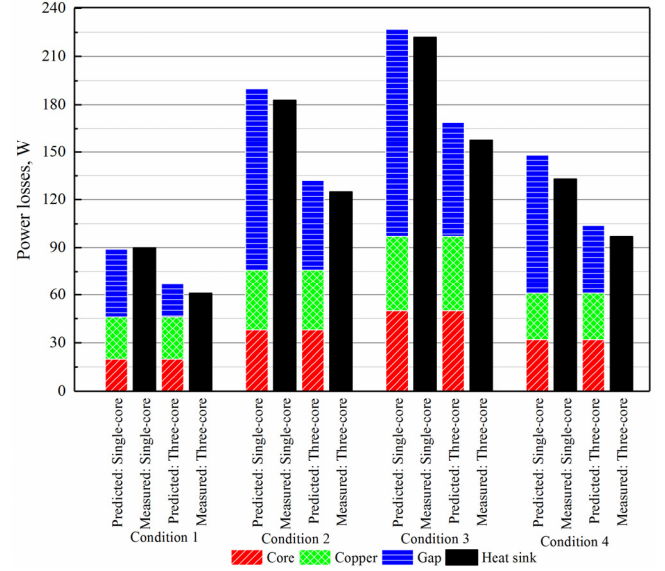


Fig. 11. Comparison of the estimated and measured losses for the single-core and split-core inductors.

Fig. 11 shows excellent correlation between the predicted and measured losses for all the cases, the average error is 6%, confirming the accuracy of the predictions. In particular the gap losses with the three split cores are reduced by 50 %, and since the gap losses account for a large proportion of the total losses, there is high confidence in the gap loss predictions themselves. The overall inductor losses are seen to be reduced by about 30 %.

### B. Three-dimensional Finite Element Steady-State Thermal Simulation

To provide further confirmation of the loss distribution in the inductor, the measured temperatures from the embedded sensors within the component, as shown in Fig. 9, were compared against predicted temperatures from 3-D FEA thermal simulations. The core and copper losses in the thermal model were assumed to be evenly distributed in the respective parts of the structure, whilst the gap loss distribution was from the 3-D FEA electromagnetic simulation. The core and foil winding were each represented in the thermal model as solid, homogenized blocks with anisotropic thermal conductivities to account for the different thermal properties along/through the winding and core lamination layers. The heat sink was modelled as a constant-temperature boundary on the bottom of the aluminum can. Convection was neglected since the component was covered with insulating material. The top and sides of the inductor can were modelled as adiabatic walls. Linear thermal conductivities were assumed in the model, which was solved using the Opera 3-D static thermal solver TEMPO/ST FE tool.

Fig. 12 compares the measured and predicted steady-state temperature rise for eight locations in the two inductors when operating under conditions 2 and 3, as defined in Table III. The measured and predicted temperatures generally correspond closely with one or two exceptions. The average error for condition 2 is 2.7% for the single core inductor, and 4.2 % for the split-core inductor, whilst for condition 3 it is



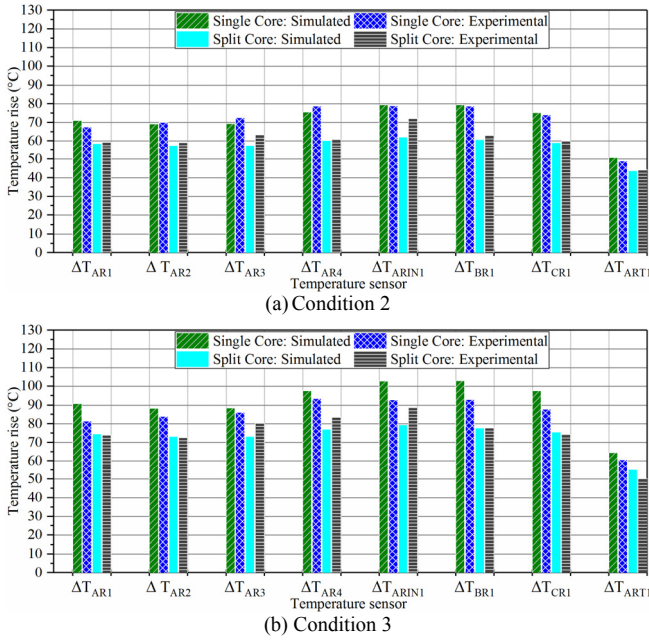


Fig. 12. Comparison of predicted steady-state and measured temperature rise, (a) Mid-gap loss condition, (b) High gap loss condition.

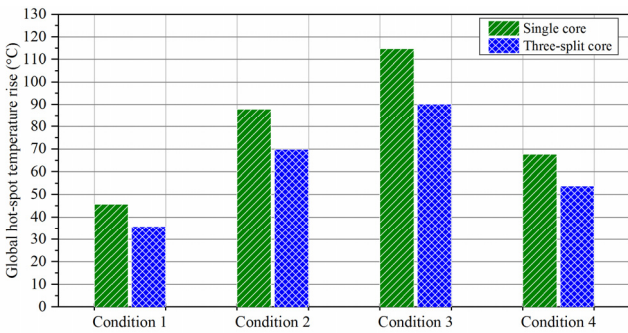


Fig. 13. Predicted FE global hot spot temperatures.

7.8 % for the single-core inductor and 5.1 % for the split-core inductor. The maximum individual errors are within 10 °C. The errors were attributed to approximations in the estimation of parameters for the thermal model and the poor thermal coupling between the split cores; no thermal interface material was used between the cores. Another potential source of error was the high temperature gradient in the core around the gap, potentially resulting in large measurement variations from small positional errors in the sensor placement. For example, considering test condition 3 then across the 2 mm × 2 mm footprint of the thermal sensor the FEA solution showed temperature differences of up to 7.4 °C for sensor ART1, 7.0 °C for sensor BR1 and 5.6 °C for sensor ARIN1.

The measurements indicate significant temperature reductions in the split core component, however, since the sensors are not necessarily positioned at exactly the hottest point in the structure, the FEA model was used to compare the global maximum core temperature between the single and split core component for the four operating conditions in Table III. The results are plotted in Fig. 13, showing a predicted reduction in the hot spot temperature of up to 24.5 °C, which implies that the split-core component could be down sized.

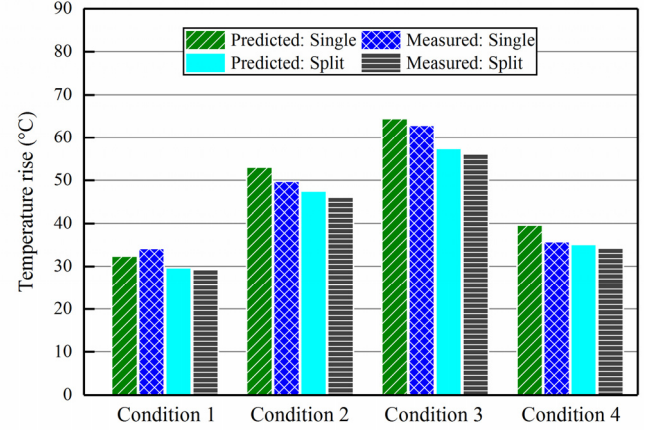


Fig. 14. Measured and predicted temperature rise inside the windings at the top of the winding.

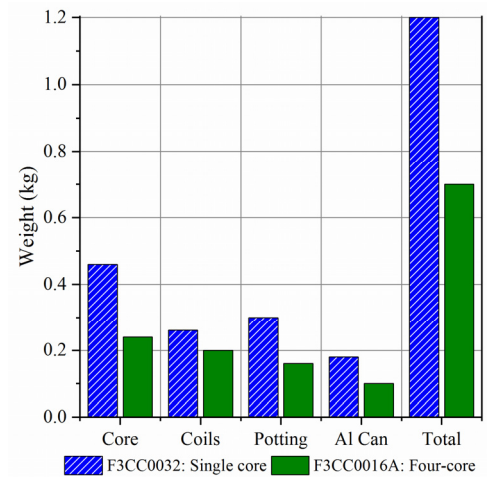


Fig. 15. Breakdown of weights for the original single-core and reduced-size inductors.

To provide further confirmation of the loss distribution within the two inductors, Fig. 14 presents measured and FEA predicted temperatures within the windings. The sensor was placed in the top of the windings, in the middle of the turn, between the second and third turns. The measurements and predictions are seen to correspond closely.

#### IV. POTENTIAL WEIGHT REDUCTION

To illustrate the possible reduction in inductor weight using a split core, the original single-core inductor was re-designed using four split cores to achieve identical inductance and current capability, that is 200 A continuous and 300 A transiently. Fig. 15 compares the weight of the original and reduced-size inductors, whilst Table IV provides design details. The new inductor comprises four sub-cores with an overall size equivalent to a F3CC0016A core, which is 45.5 % lighter than the original, F3CC0032. The overall potted weight is reduced by 0.46 kg or 42 %.

The breakdown of losses is shown in Fig. 16. The overall losses are reduced by 22.2 %. The gap losses are reduced as expected. The core losses are lower due to the significantly smaller core volume and since the AC flux is only slightly increased (by 7 %), but the copper losses are higher due to the reduced winding width and increased number of turns.



TABLE IV  
FEATURES OF THE SINGLE-CORE AND FOUR-CORE INDUCTORS

| Parameter, symbol (unit)                             | Single-core | Split-core |
|------------------------------------------------------|-------------|------------|
| Number of cores, $n_c$ (-)                           | 1           | 4          |
| Strip width, $D$ (mm)                                | 29.9        | 6.4        |
| Thickness of core limbs, $E$ (mm)                    | 13.3        | 11.0       |
| Length of core limbs, $F$ (mm)                       | 28.1        | 20.0       |
| Number of turns, $N$ (-)                             | 6           | 8          |
| Overall gap length, $l_g$ (mm)                       | 4.0         | 6.3        |
| Copper foil cross-sectional area, (mm <sup>2</sup> ) | 30.0        | 22.9       |

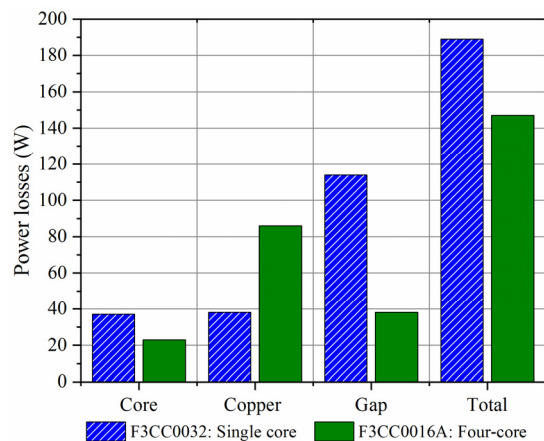


Fig. 16. Breakdown of power losses for the original single-core and reduced-size inductors for operating condition 2.

FEA thermal analysis of the new inductor confirmed the feasibility of the design; the maximum steady-state core temperature was within the material limit of 155 °C when the cold plate temperature was 60 °C.

## V. CONCLUSION

Through extensive FEA modelling the gap losses in nanocrystalline cores are seen to increase non-linearly with core width, in particular becoming very low when the core width is around the effective skin depth in the core. Over the frequency range 10 kHz – 200 kHz the gap losses are approximately proportional to  $D^\alpha$ , where  $D$  is the core width and is a  $\alpha$  constant,  $1.6 < \alpha < 1.9$ , however  $\alpha$  depends on the gap length, increasing with the gap. These relations are seen to hold true over the range of core dimensions typically encountered in power electronic converters.

As a result of the non-linear dependence of gap loss on core width, the gap loss may be significantly reduced by splitting the core into several sub-cores placed side-by-side. Finite element simulations and experimental results have demonstrated that with three sub-cores and a gap length of 4 mm, the gap losses are approximately halved in a 300 A (peak), 200 A (continuous), 150 kHz inductor. Greater reductions may be achieved in inductor designs with larger gap lengths and by using a greater number of split cores. However, at frequencies below 10 kHz and with narrower cores, the benefits of split cores will be lower. Whilst the technique has been demonstrated using Finemet nanocrystalline cores, similar benefits are likely to be found with other tape-wound cores. Compared to the usual technique of placing multiple gaps within the magnetic circuit to lower the gap losses, this method is likely to be easier to fabricate

since the ribbon may be cut to width before the core is wound and avoids the problems associated with cutting the fine core laminations and possibly creating short-circuits between them. The increased cost of customized core sizes is a disadvantage of this approach; however the cost should reduce significantly for volume manufacturing.

By providing a simple technique to reduce gap losses, which can be the dominant loss in some high current inductors, this paper offers one route to reduce inductor size when the design is thermally limited. A case study has shown that with four split cores the weight of the original inductor considered in the paper could be reduced by 0.5 kg or 42 %. Assuming that the magnetic components account for 40 % of a converter's total weight, the proposed technique could easily reduce the converter weight by 16 %. The overall reduction is likely to be greater due to additional benefits such as a reduced heat sink area and smaller case size.

The design results in this paper also illustrate that achieving further size reductions in high current inductors requires lower winding losses, for example using techniques to minimize the AC resistance, and also improved thermal management such as embedded heat paths in the high loss regions of the structure.

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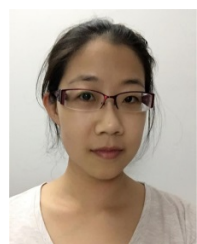
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**Gerardo Calderon-Lopez** received the degree in Ing. en Comunicaciones y Electrónica from the National Polytechnic Institute (IPN), Mexico City in 1999, the M.Sc. degree in Power Electronics and Drives from The Universities of Birmingham and Nottingham, U.K. in 2001, and the Ph. D. degree in Power Electronics from The University of Manchester, U.K. in 2009.

He is a Research Associate at The University of Manchester. His research interests include power-dense converters and magnetic components.



**Yiren Wang** received the B.Eng.(Hons.) degree in Electrical and Electronic Engineering from The University of Manchester, Manchester, U.K. and North China Electric Power University, Baoding, China, in 2011, and the Ph.D. degree in Power Electronics from The University of Manchester, Manchester, U.K., in 2015.

She was a Research Associate at The University of Manchester from 2016 to 2018. She is currently a Power Electronic Design Engineer at ZyteK Automotive/Continental Engineering Services, U.K.

Her research interests include magnetic components for high power density DC-DC converters, modeling, design and optimisation.



**Andrew J. Forsyth** (M'98) received the B.Sc.(Eng) degree in Electrical Engineering from Imperial College, London, U.K., in 1981 and the Ph.D. degree in Power Electronics from the University of Cambridge, Cambridge, U.K., in 1987.

He was a Design Engineer with GEC Electrical Projects Ltd from 1981 to 1983, a Lecturer at the University of Bath from 1986-1990, and a Lecturer / Senior Lecturer at Birmingham University from 1991 to 2004. Since 2004 he has been Professor of Power Electronics at the University of Manchester,

Manchester U.K. His research interests include high-frequency converters and magnetic components, converter modelling and control, aerospace and electric vehicle applications.